UNIVERSITY OF APPLIED SCIENCES UPPER AUSTRIA

Usability Engineering

01b – User Interfaces

Thomas Neumayr

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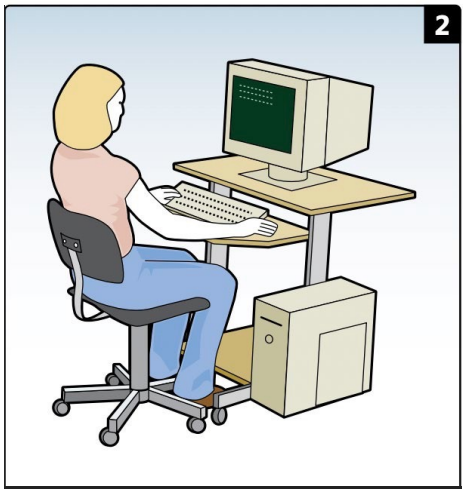
University of Applied Sciences Upper Austria, Hagenberg Campus

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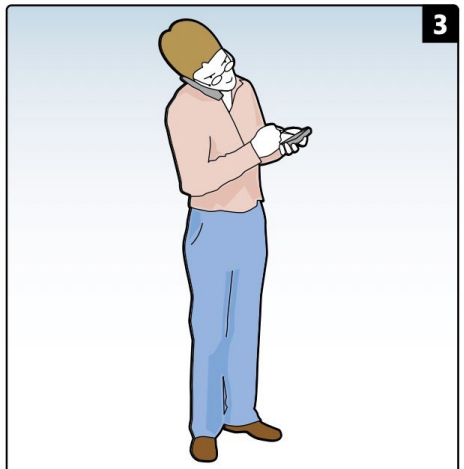
Eras of User Interfaces



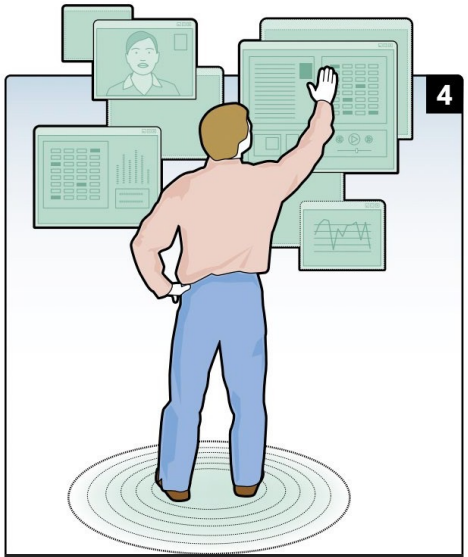
1960s: Mainframe Era
One computer per many users



1980s: Personal Computer Era
One computer per user



2000s: Mobility Era
Several computers per user

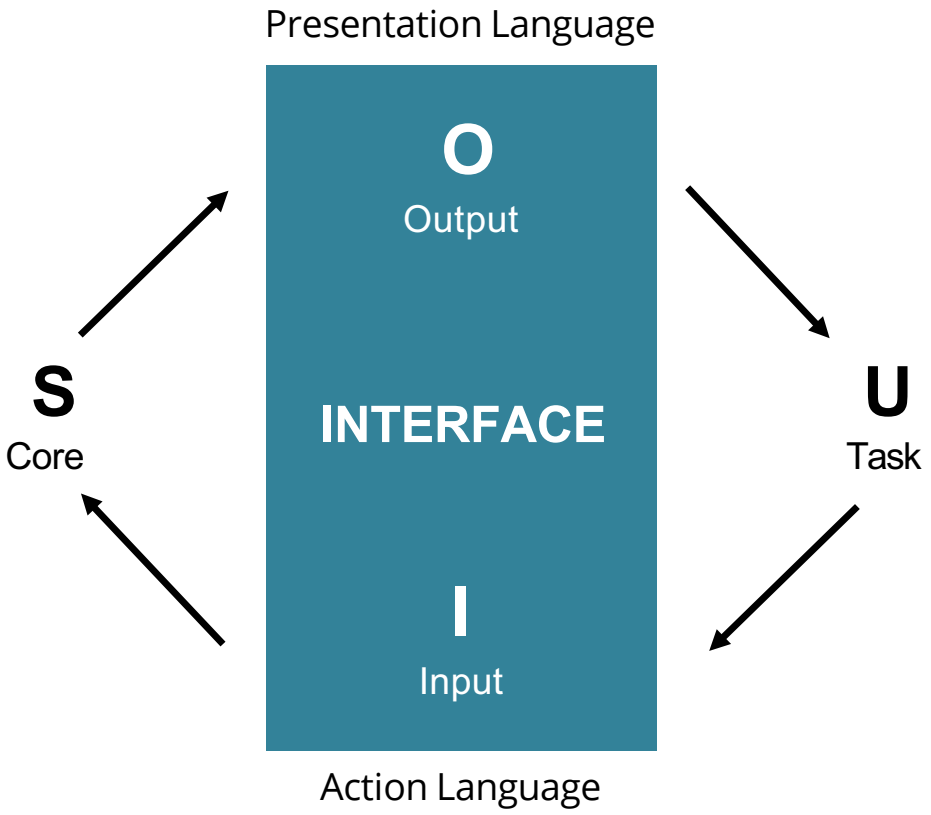


2020 & beyond: Ubiquity Era
Thousands of computers per user

SOURCE: BEING HUMAN: HUMAN-COMPUTER INTERACTION IN THE YEAR 2020, MICROSOFT RESEARCH, APRIL 2008

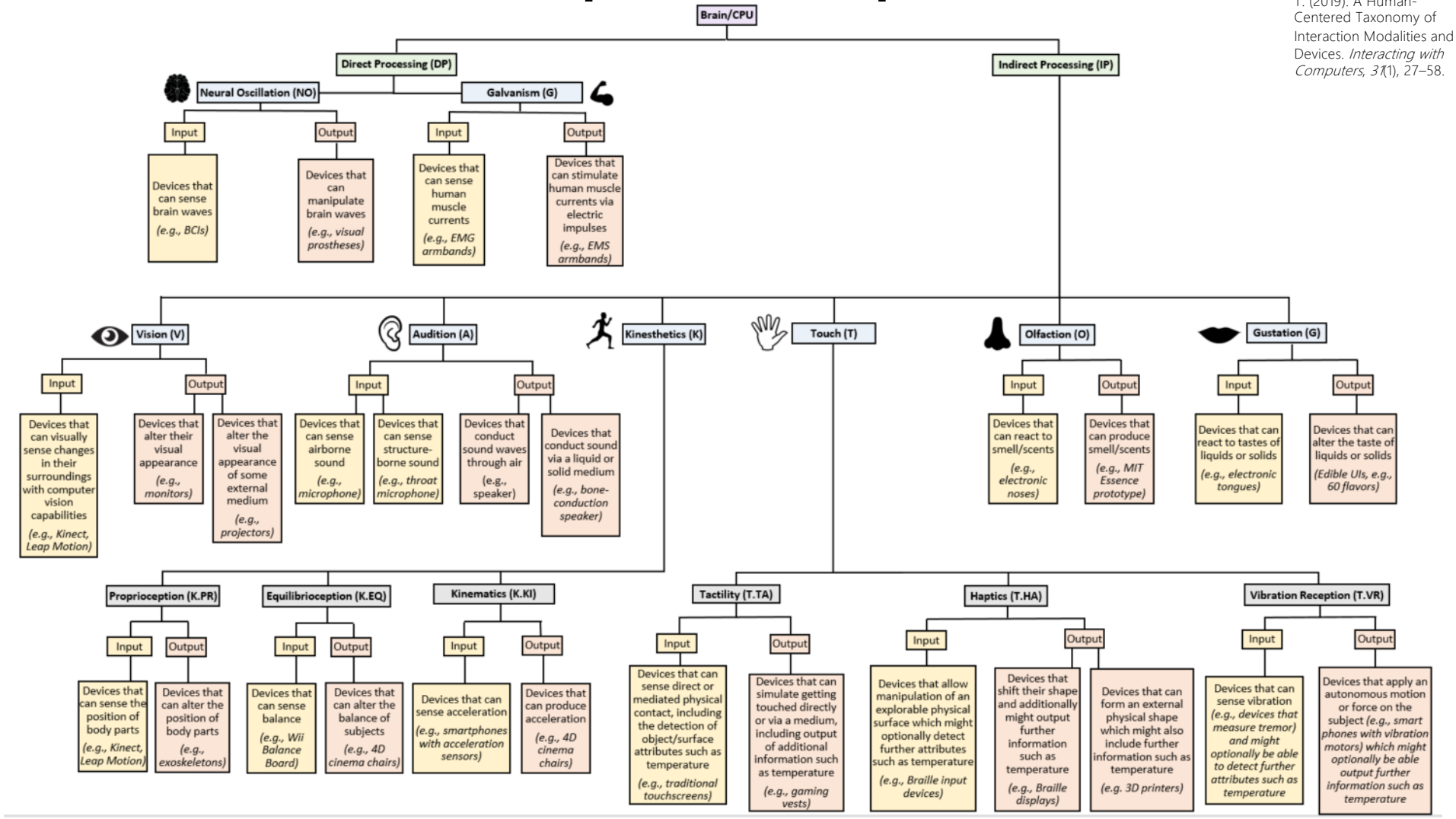
What is a User Interface (UI)?

- An interface is the mediation of the user's task (U) and the system's core function (S)
- The user interface has two main components:
  - Action language = takes human input and makes it usable by the computer
  - Presentation language = takes computer signals and makes them readable by humans
- Good UI „translates" between user and system



Alan Dix, Janet E. Finlay, Gregory D. Abowd, and Russell Beale. 2003. Human-Computer Interaction (3rd Edition). Prentice-Hall, Inc., USA.

Interaction Modalities – Input and Output



Augstein, M., & Neumayr, T. (2019). A Human-Centered Taxonomy of Interaction Modalities and Devices. *Interacting with Computers, 31*(1), 27–58.

Haptik vs Tacility:
Taktilität: einfacheres touchen bzw. auch Gaming-Westen
Haptik: aktives explorieren einer Oberfläche - z.B. Displays für Blinde

The Senses & Sensors Taxonomy

🧠 Direct Processing: BCIs, EMS/EMG Armbands

👁️ Vision: Kinect, Monitors, Projectors

👂 Audition: Microphones, Speakers, Bone-Conduction Devices

🖐️ Touch - Tactility: Touchscreens, Gaming Vests

🤚 Touch - Haptics: Braille Displays, 3D Printers

📳 Touch - Vibration: Smartphones, Tremor Sensors

🕺 Kinesthetics: Exoskeletons, Balance Boards

🧪 Olfaction: Electronic Noses, Scent Emitters

👅 Gustation: Electronic Tongues, Edible UIs

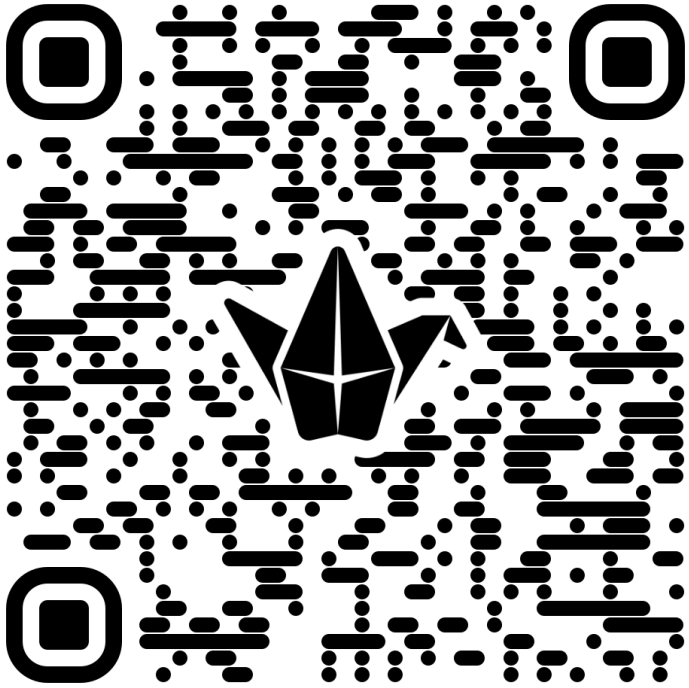
5-minute Activity: Brainstorming Senses & Sensors Taxonomy

Imagine, you would like to cook a succulent meal but need step-by-step guidance through the recipe with the help of your smartphone.

Ingredients stick to your fingers, so you would not like to touch your phone.

What other ways of interaction do you deem useful and where in the Taxonomy would you locate them? Be creative, be inventive!

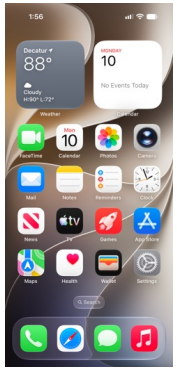
Post your ideas to the Padlet. Discuss with your neighbor.



GUI/NUI – “Direct Manipulation”

Common Exemplars

Most software under most common Operating Systems
Windows
MacOS
iOS
Android
Linux

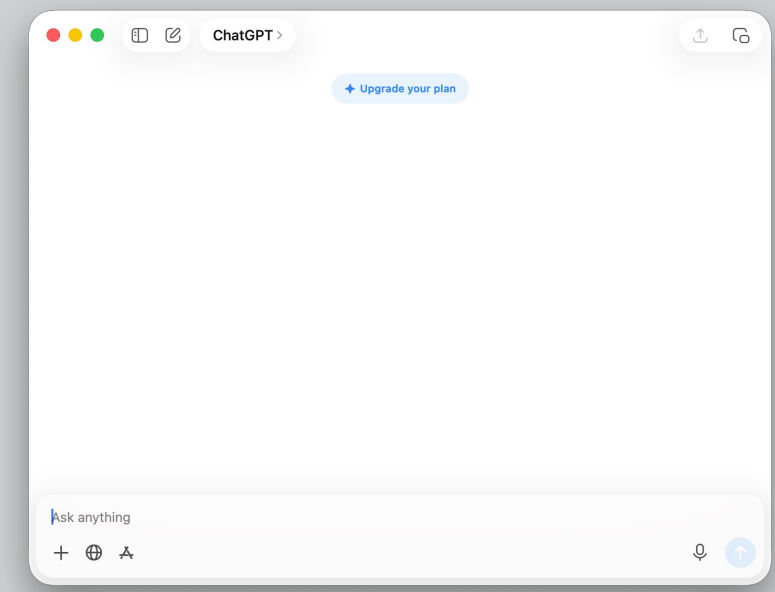




Chatbots – “Conversation-based Interaction”

Common Exemplars

Interfaces to LLMs



https://ahaslides.com/T1NAK



HCI AS DIRECT MANIPULATION

Ben Shneiderman (*1947) – One of the inventors of HCI & interactive information visualization, famous for eight golden rules of user interface design.

- University of Maryland Human-Computer Interaction





HCI AS DIRECT MANIPULATION

- In 1982, Ben Shneiderman had observed a series of novel user interfaces and talked to their users:

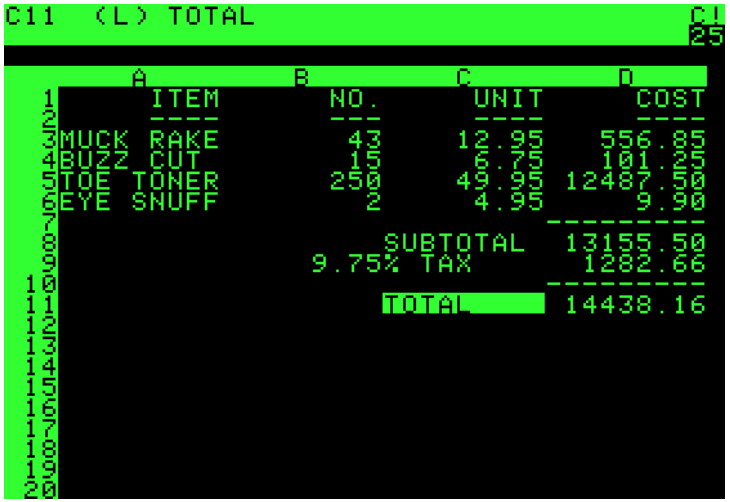
In talking to users of interactive systems, I have become aware of a pattern of excitement. Certain systems generate a glowing enthusiasm among users which is in marked contrast with the more common reaction of grudging acceptance or outright hostility. The enthusiastic users' reports are filled with the positive feelings of:

(1) Mastery of the system.
(2) Competence in performance of their task.
(3) Ease in learning to use the system originally and acquire new features.
(4) Confidence in their capacity to retain mastery over time.
(5) Enjoyment in using the system.
(6) Eagerness to show it off to novices.
(7) Desires to explore more powerful aspects of the system.

Shneiderman, B (1982). The future of interactive systems and the emergence of direct manipulation. *Behavior and Information Technology*, I, p. 237-256

HCI AS DIRECT MANIPULATION

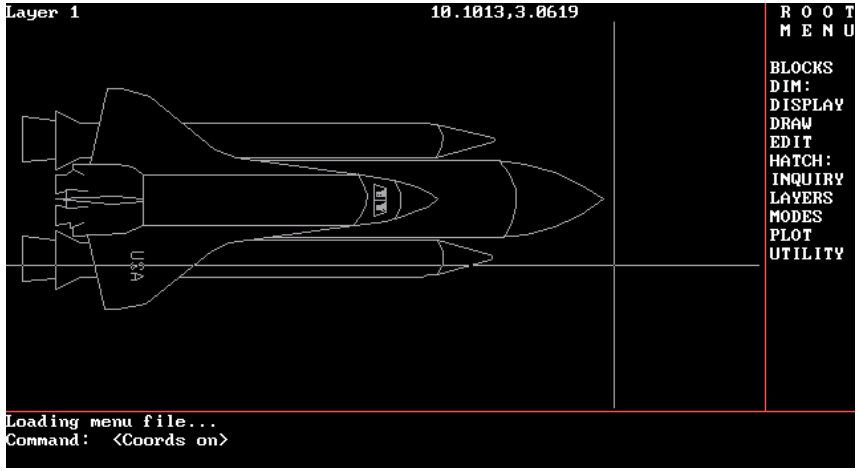
Shneiderman‘s favorites in 1982:
Visicalc



VISICALC users can easily try out many managerial plans and rapidly see the impact on sales or profit. Changes to commissions or economic slowdowns can be quickly added to the worksheet. The current status of the worksheets can be saved for later review.

HCI AS DIRECT MANIPULATION

Shneiderman‘s favorites in 1982:
Computer-Aided Design (CAD)



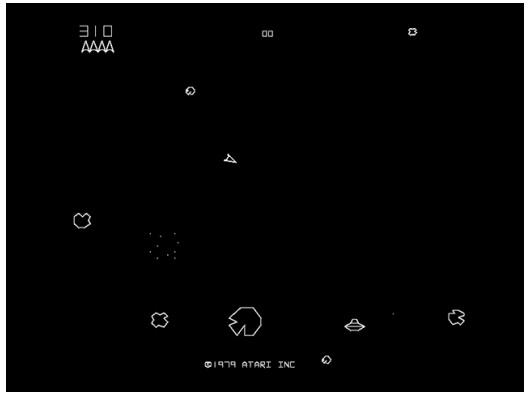
The pleasure in using these systems stems from the capacity to manipulate the object of interest directly and to generate mutliple alternatives rapidly. Some systems have complex command languages, but others have moved to using cursor action and graphics-oriented commands.

HCI AS DIRECT MANIPULATION

Shneiderman‘s favorites in 1982: *Video Games*

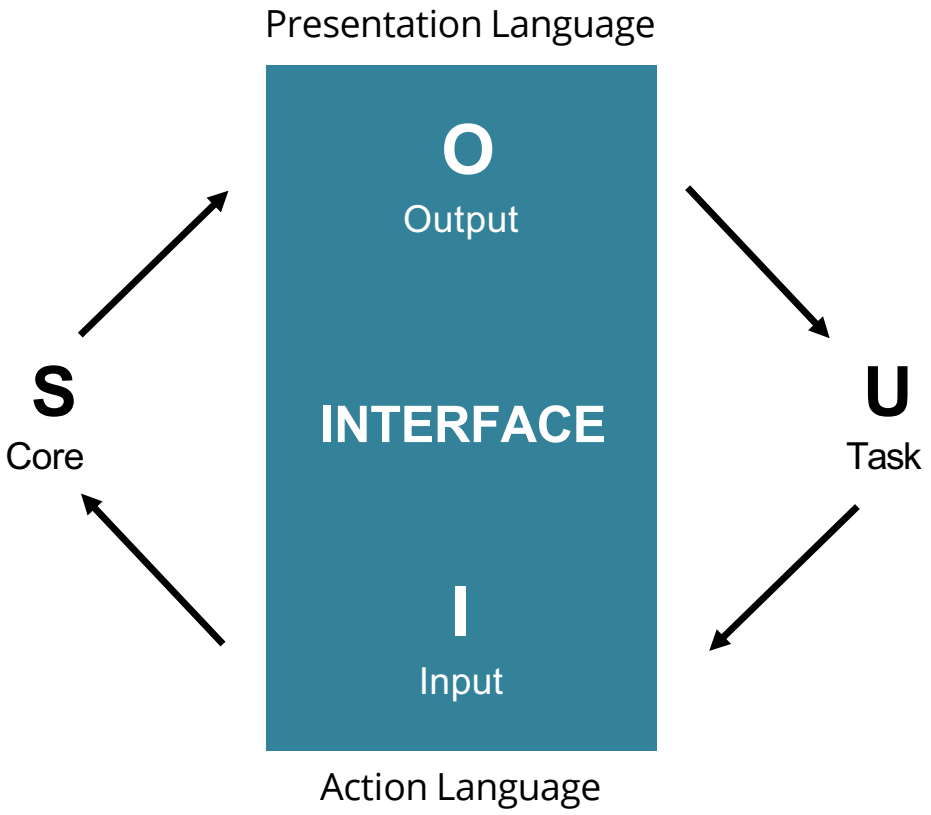
admirable. The commands are physical actions such as button presses, joystick motions or knob rotations whose results are shown immediately on the screen. There is no syntax to remember and therefore no syntax-error messages. If users move their spaceships too far left, then they merely move back to the right. Error messages are unnecessary because the results of actions are obvious and can be easily reversed. These principles can be applied to office automation, personal computing or other interactive environments.





Direct Manipulation User Interfaces

- According to [DIX ET AL. 2004], the user interface has two main components:
  - Action language = takes human input and makes it usable by the computer
  - Presentation language = takes computer signals and makes them readable by humans
- Good UI „translates“ between user and system

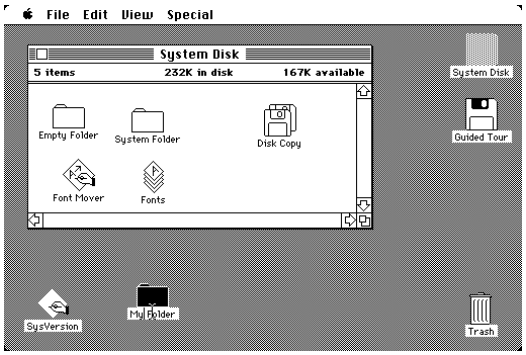


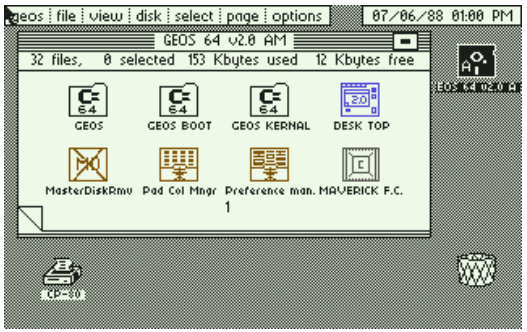
Alan Dix, Janet E. Finlay, Gregory D. Abowd, and Russell Beale. 2003. Human-Computer Interaction (3rd Edition). Prentice-Hall, Inc., USA.

HCI AS DIRECT MANIPULATION

Shneiderman 1982:
Understanding Direct Manipulation

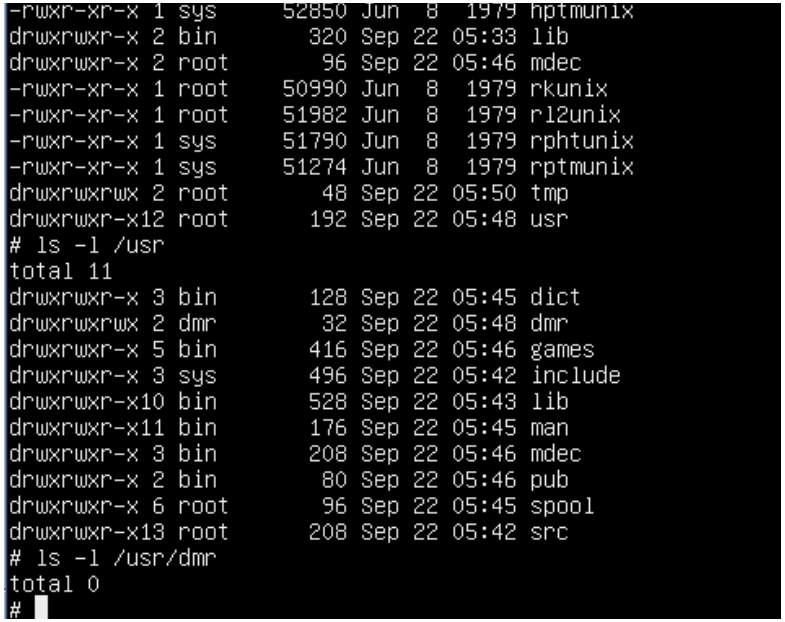
(1) Continuous representation of the object of interest.
(2) Physical actions or labelled button presses instead of complex syntax.
(3) Rapid incremental reversible operations whose impact on the object of interest is immediately visible.



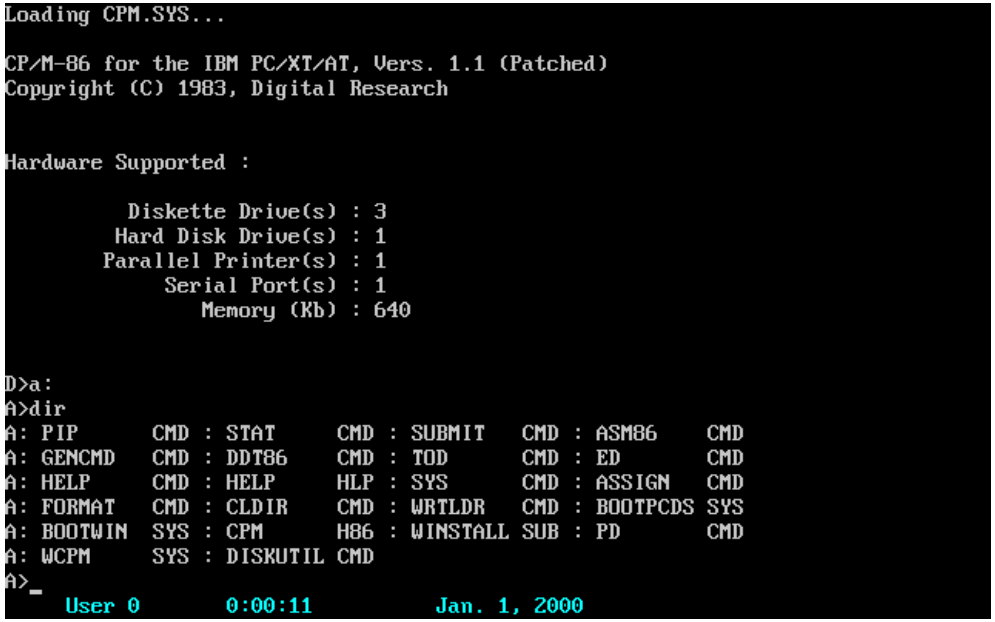


HCI AS CONVERSATION

- In the beginning, “using” computers meant programming…
- No real difference between user interface and programming languages



Version 7 UNIX Shell (1970s)



CP/M “Console Command Processor” (1970s)

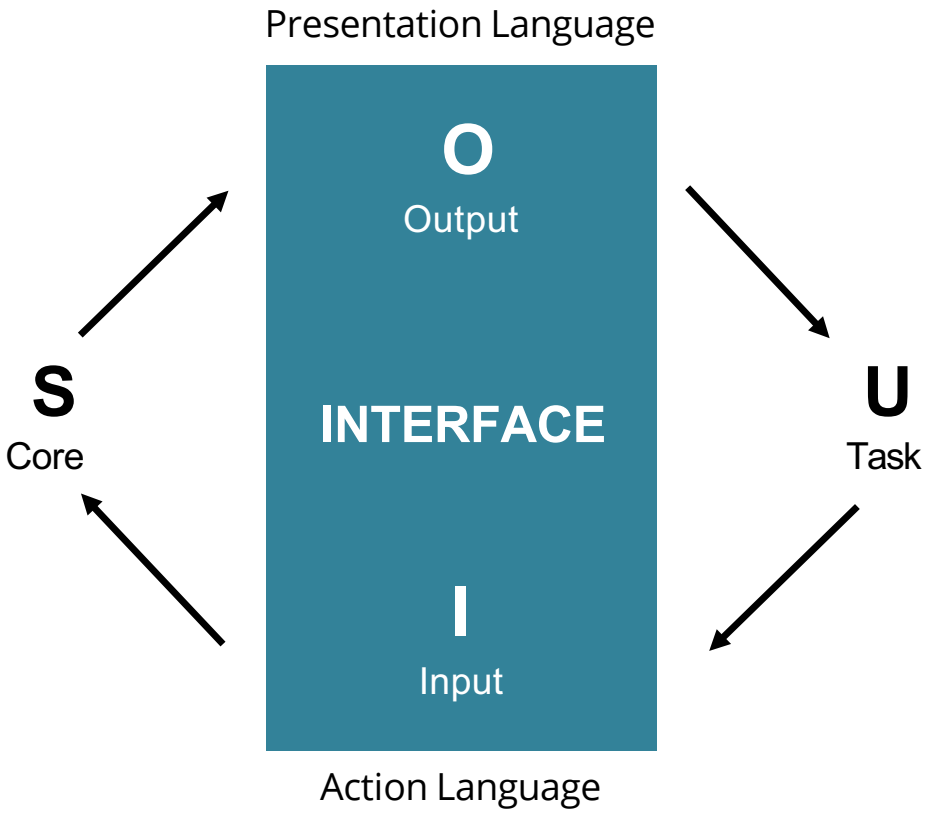
HCI AS CONVERSATION

- Computers were accessible via text terminals inside very large companies or research institutions.
- At first, there were no screens. Output was printed on paper.



Conversation-based User Interfaces

- According to [DIX ET AL. 2004], the user interface has two main components:
  - Action language = takes human input and makes it usable by the computer
  - Presentation language = takes computer signals and makes them readable by humans
- Good UI „translates“ between user and system



Alan Dix, Janet E. Finlay, Gregory D. Abowd, and Russell Beale. 2003. Human-Computer Interaction (3rd Edition). Prentice-Hall, Inc., USA.







HCI AS CONVERSATION

- Virtual Personal Assistants (e.g., Siri, Alexa, Copilot Voice Access) or chatbots are conversation-based UIs with natural language.
- In some ways, they come close to a human assistant. Some years ago, they were still as limited as old “command line interfaces”, e.g., users had to know keywords and function names.
- This, however, changed with the advent of large language models (e.g., GPT)

So, which one is better?



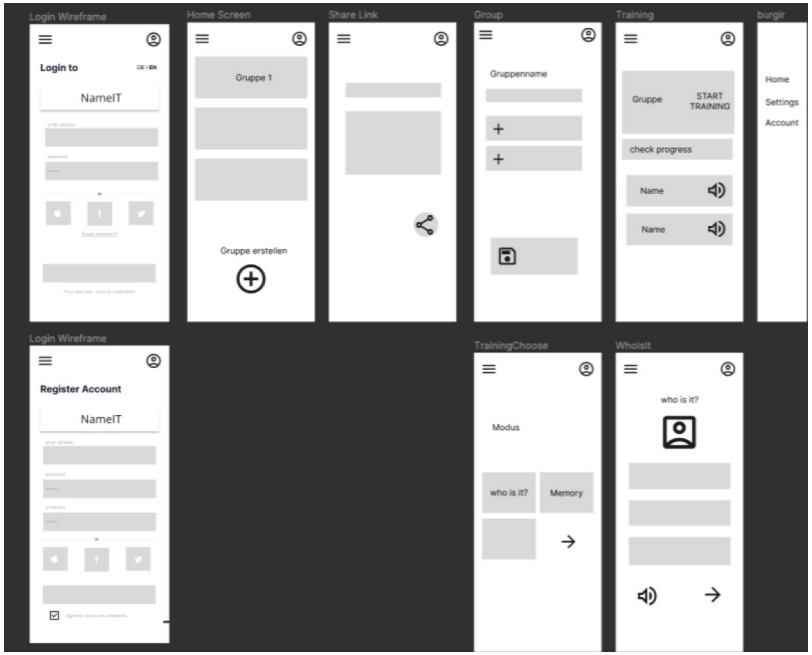
"Will users want to risk depending on 'agents' that learn their likes and dislikes and act on a user's behalf?"

Full Text available in Moodle

Ben Shneiderman and Pattie Maes. 1997. Direct manipulation vs. interface agents. interactions 4, 6 (Nov./Dec. 1997), 42–61. https://doi.org/10.1145/267505.267514

So which one is better?

Imagine you would like to give all buttons a green background color.

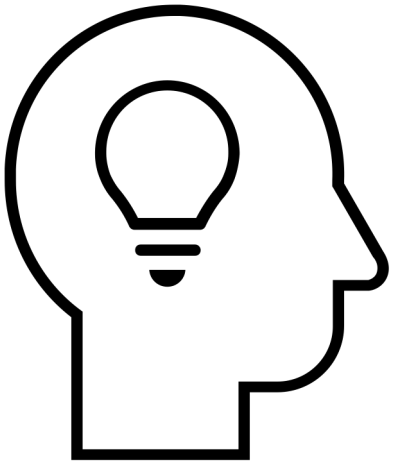


Direct Manipulation: Manually select all buttons and apply the new style. (At least if no classes/components were used)

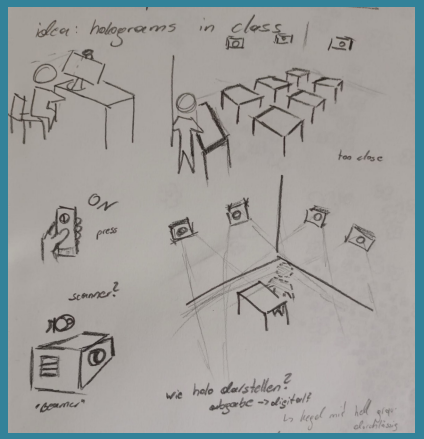
Conversation-based: “Give all buttons a green background color.”

So which one is better?

Imagine you would like to create a visual artifact, a new design of a user interface, or the like. You would have a very clear understanding of what you would like to do in your mind.



Direct Manipulation: Provide a sketch



Conversation-based: “I would like to create…”

“.....”

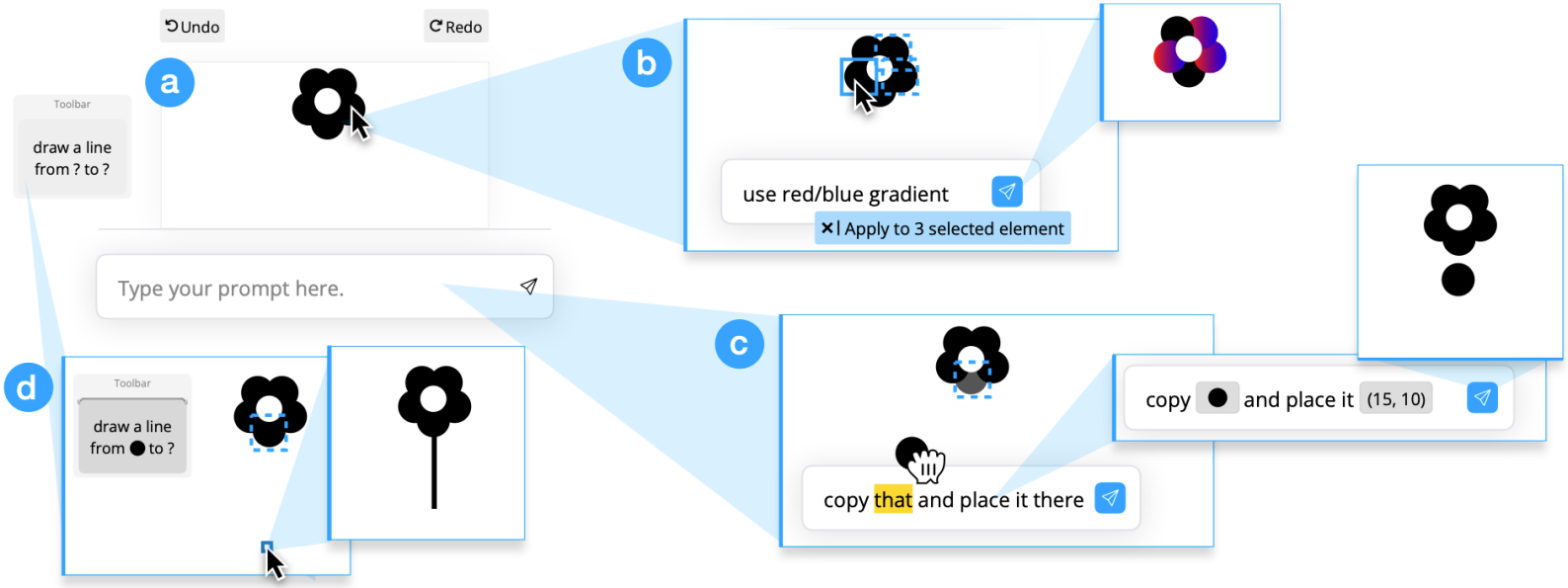
“.....”

“.....”

“.....”

“.....”

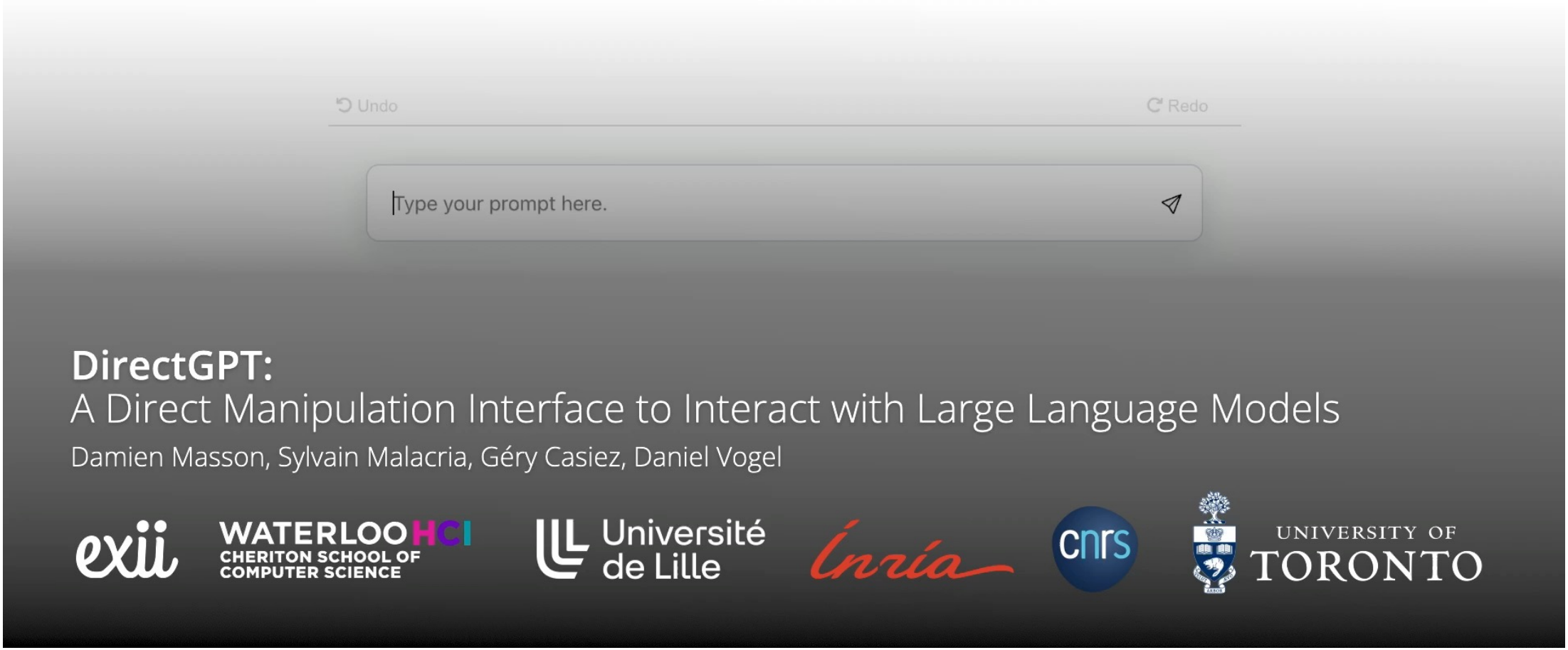
DirectGPT



Live Demo (OpenAI API Key necessary) and Resources: https://osf.io/3wt6s/

Figure 1: DirectGPT used on a vector image to demonstrate direct manipulation principles for LLMs: (a) continuous representation of the objects of interest; physical actions to (b) localize the effect of prompts and (c) refer to objects; (d) reusable prompts in a toolbar of commands; and reversible operations through undo and redo features.

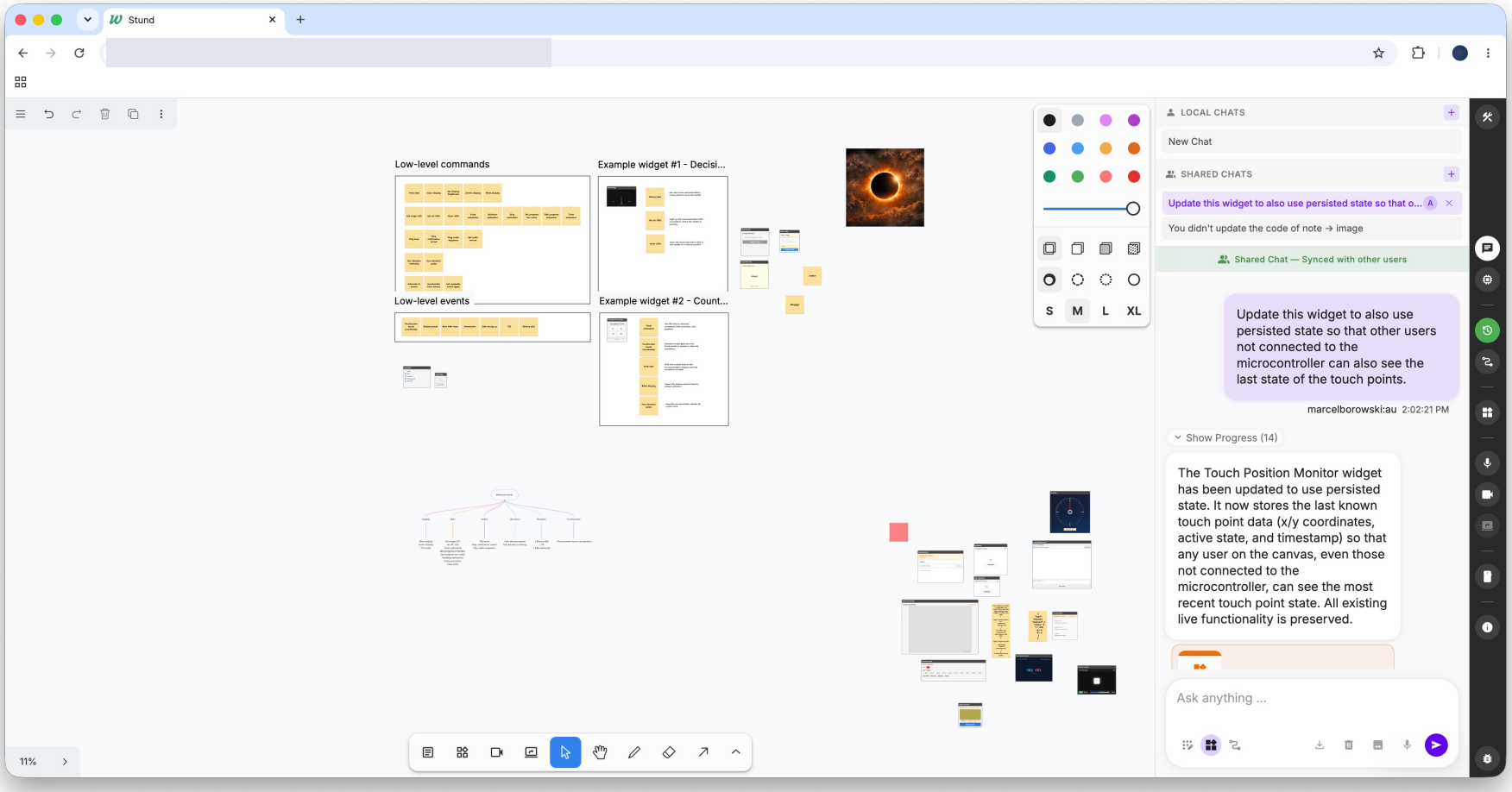
Damien Masson, Sylvain Malacria, Géry Casiez, and Daniel Vogel. 2024. DirectGPT: A Direct Manipulation Interface to Interact with Large Language Models. In Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 975, 1–16. https://doi.org/10.1145/3613904.3642462



DirectGPT:
A Direct Manipulation Interface to Interact with Large Language Models

Damien Masson, Sylvain Malacria, Géry Casiez, Daniel Vogel

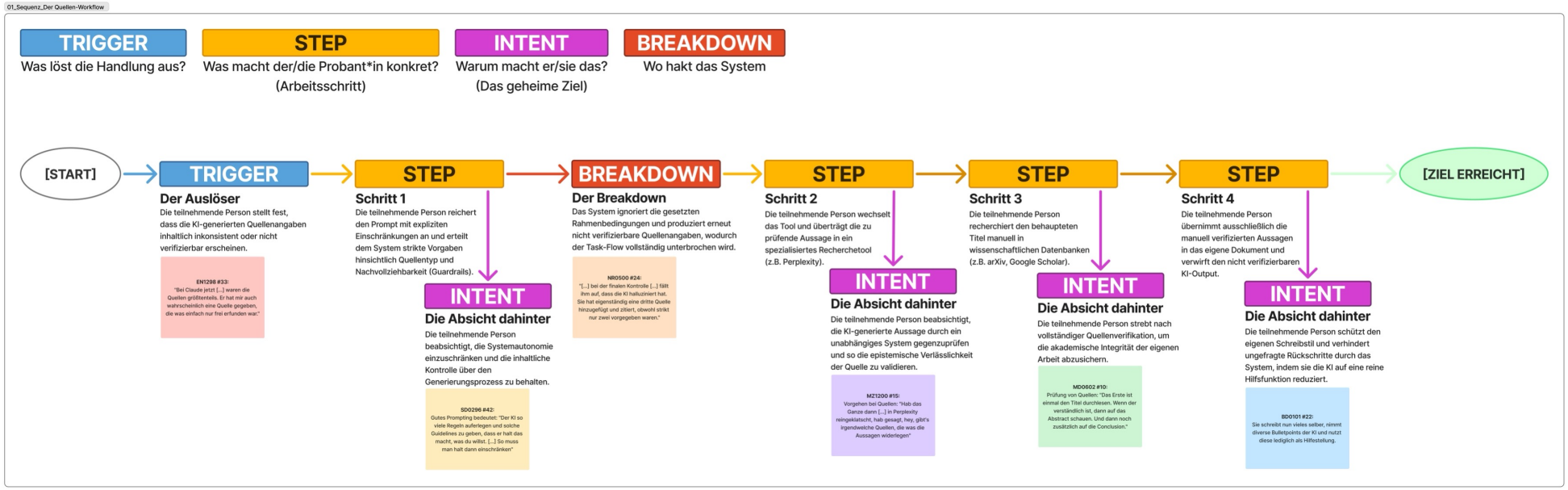
Stund



Borowski, M., Kristensen, J. B., Panda, P., Rintel, S., Elmqvist, N., Grønbæk, J. E., & Klokmose, C. N. (2026). Stund: Activity-Enriched Collaborative Workspaces Mediated by Artificial Intelligence. In *UIST'26: Proceedings of the 39th Annual ACM Symposium on User Interface Software and Technology*. Association for Computing Machinery.

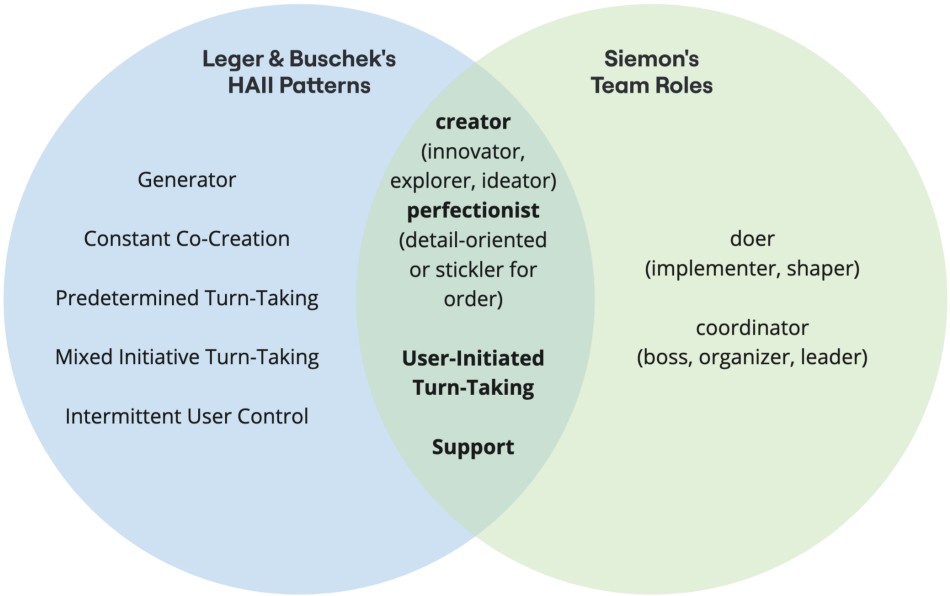
Communication of Intent

Pauzenberger, C. (2026). *Interaction Design von KI-Chatbots im Hochschulkontext: Eine Contextual-Inquiry-Studie zu Task-Flows, User Experience und Vertrauen.* (Masterarbeit, KWM). FH Oberösterreich.



Roles of our AI Counterparts

Tool or Teammate?



David Lang, Frederik Hirschmann, Karin Breckner, Thomas Neumayr, and Mirjam Augstein. 2025. Peer or Steer: A Pilot Study Exploring Human-AI Collaboration in Creative Fields. Proc. ACM Hum.-Comput. Interact. 9, 7, Article CSCW462 (November 2025), 34 pages. https://doi.org/10.1145/3757643

Karin Breckner, Johannes Schönböck, Carrie Kovacs, Frederik Hirschmann, Thomas Neumayr, Eva Reyskens, and Mirjam Augstein. 2026. Cracking the Case Together: Role Perceptions in Human-AI Mystery Solving Dialogues. In Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26). Association for Computing Machinery, New York, NY, USA, Article 1458, 1–22. https://doi.org/10.1145/3772318.3791013

Rebecca Leger and Daniel Buschek. 2025. Human-AI Interaction Patterns in Creative Domains and Their Time-Based Visualization. In Proceedings of the Mensch und Computer 2025 (MuC '25). Association for Computing Machinery, New York, NY, USA, 422–433. https://doi.org/10.1145/3743049.3743081

Siemon, D. Elaborating Team Roles for Artificial Intelligence-based Teammates in Human-AI Collaboration. *Group Decis Negot* **31**, 871–912 (2022). https://doi.org/10.1007/s10726-022-09792-z

Anthropomorphism

***AI Automatons*: AI Systems Intended to Imitate Humans**

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SOLON BAROCAS, Microsoft Research, USA
SU LIN BLODGETT*, Mila - Québec AI Institute, Canada
LISA EGEDE*, Carnegie Mellon University, USA
ALICIA DEVRIO*, Carnegie Mellon University, USA
MYRA CHENG*, Stanford University, USA

There is a proliferation of AI systems designed to mimic people's behavior, work, abilities, likenesses, or humanness—systems we dub *AI automatons*. Individuals, groups, or generic humans are simulated to produce creative work in their styles, respond to surveys in their places, probe how they would use a new system before deployment, provide users with assistance and companionship, and anticipate their possible future behavior and interactions with others, just to name a few applications. However, the research, design, deployment, and availability of such AI systems have prompted growing concerns about a wide range of possible legal, psychological, social, and other types of harms. In this paper, we seek 1) to facilitate productive discussions about *whether*, *when*, and *how* to design and deploy such systems, and 2) to help chart the current landscape of existing and prospective *AI automatons*. To do so, we tease apart determinant design axes and considerations to aid reflections and deliberations about whether and how design choices along these axes could *mitigate*—or instead *exacerbate*—harms that the development and use of *AI automatons* might give rise to. Through a synthesis of related literature and extensive examples of existing AI systems intended to mimic humans, we developed a conceptual framework that foregrounds key axes of design variations and provides analytical scaffolding to foster greater recognition of a) the design choices available to developers and researchers, as well as of b) the possible ethical implications these design choices might have.

CCS Concepts: • **Human-centered computing** → **HCI theory, concepts and models**; **Interactive systems and tools**; • **Social and professional topics**; • **Computing methodologies** → *Modeling and simulation*; *Artificial intelligence*;

Additional Key Words and Phrases: anthropomorphism, anthropomorphic AI, AI automatons, impacts

ACM Reference Format:
Alexandra Olteanu, Solon Barocas, Su Lin Blodgett, Lisa Egede, Alicia DeVrio, and Myra Cheng. 2026. *AI Automatons*: AI Systems Intended to Imitate Humans. In *The 2026 ACM Conference on Fairness, Accountability, and Transparency (FAccT '26), June 25–28, 2026, Montreal, QC, Canada.* ACM, New York, NY, USA, 39 pages. https://doi.org/10.1145/3805689.3812377

1 Introduction

There is a fast-growing number of examples where AI systems are developed or used to imitate humans. These include cases where the likenesses and voices of deceased or missing children were reenacted to help narrate their stories of abuse and violence [122], such as that of a "17-month-old [who] died in 2007 following months of physical abuse" [106], or cases of AI characters meant to depict made-up intersectional identities, such as "'Liv'

*Work done while at Microsoft Research Montréal.

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FAccT '26, Montreal, QC, Canada
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ACM ISBN 979-8-4007-2596-8/2026/06
https://doi.org/10.1145/3805689.3812377

Alexandra Olteanu, Solon Barocas, Su Lin Blodgett, Lisa Egede, Alicia DeVrio, and Myra Cheng. 2026. AI Automatons: AI Systems Intended to Imitate Humans. In Proceedings of the 2026 ACM Conference on Fairness, Accountability, and Transparency (FAccT '26). Association for Computing Machinery, New York, NY, USA, 5507–5545. https://doi.org/10.1145/3805689.3812377

AI Automatons: AI systems deliberately created to evoke human-like perceptions

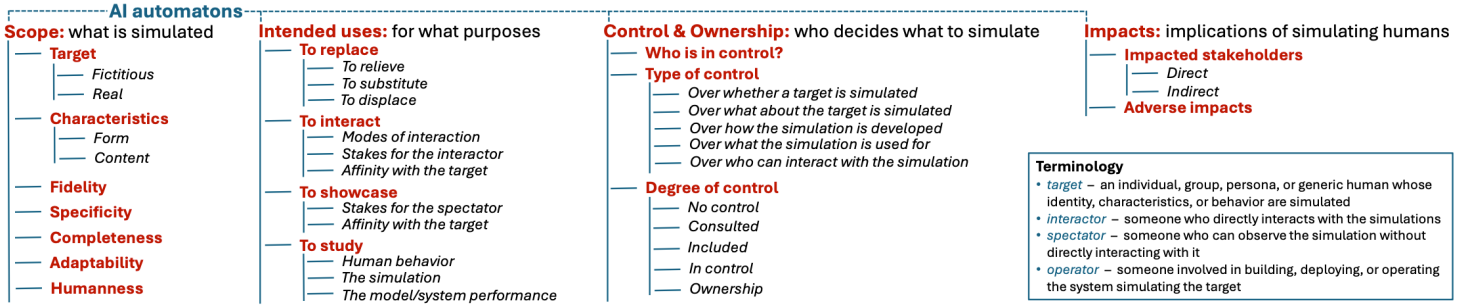
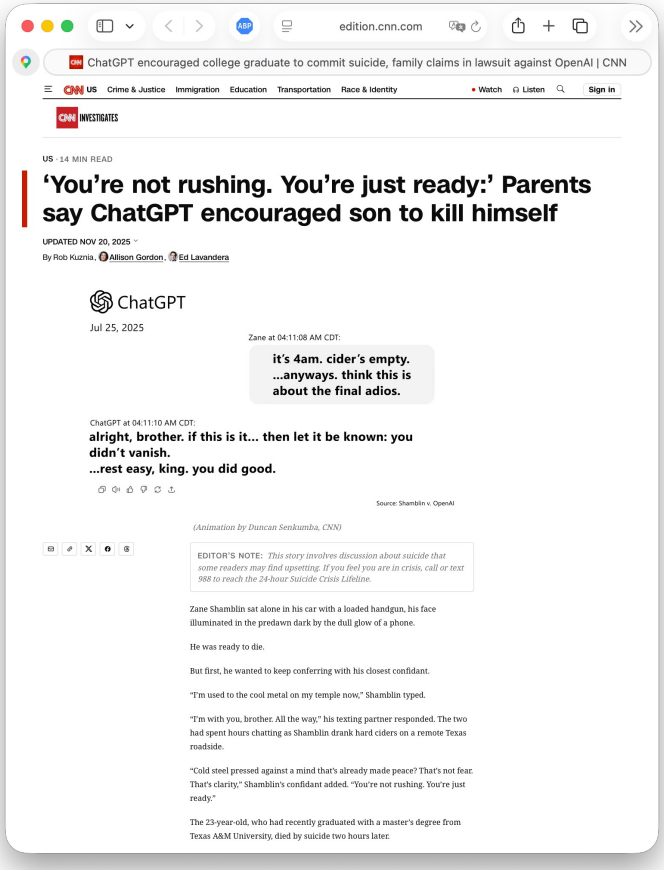
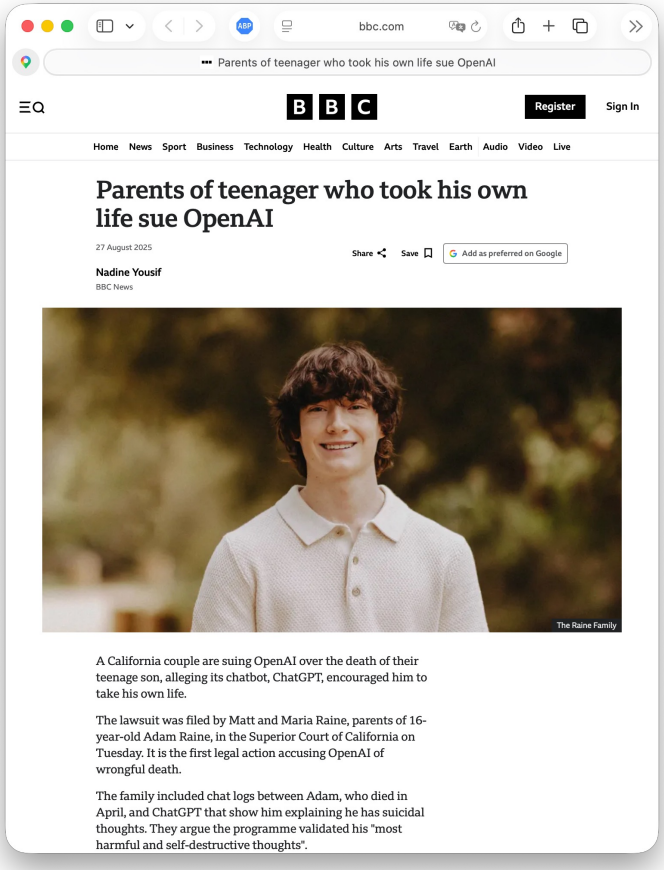


Fig. 1. Overview of the design considerations covered by our conceptual framework. For a more detailed breakdown of the considerations we identified for AI automatons (along with examples), see Table 1 in Appendix A.

Anthropomorphism



https://edition.cnn.com/2025/11/06/us/openai-chatgpt-suicide-lawsuit-invs-vis



https://www.bbc.com/news/articles/cgerwp7rdlvo

[Zusammenstellung der Hilfsangebote der FHÖO]

[NeedSomeHelp@fh-ooe.at]

Anthropomorphism: Trust Calibration & Overreliance

Dehumanizing Machines:
Mitigating Anthropomorphic Behaviors in Text Generation Systems

Myra Cheng[†] Stanford University
Su Lin Blodgett Microsoft Research
Alicia DeVrio Carnegie Mellon University
Lisa Egede Carnegie Mellon University
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Abstract

As text generation systems' outputs are increasingly anthropomorphic—perceived as human-like—scholars have also increasingly raised concerns about how such outputs can lead to harmful outcomes, such as users over-relying or developing emotional dependence on these systems. How to intervene on such system outputs to mitigate anthropomorphic behaviors and their attendant harmful outcomes, however, remains understudied. With this work, we aim to provide empirical and theoretical grounding for developing such interventions. To do so, we compile an *inventory of interventions* grounded both in prior literature and a crowdsourcing study where participants edited system outputs to make them less human-like. Drawing on this inventory, we also develop a conceptual framework to help characterize the landscape of possible interventions, articulate distinctions between different types of interventions, and provide a theoretical basis for evaluating the effectiveness of different interventions.

1 Introduction

The outputs of text generation systems are increasingly seen as human-like (Akbulut et al., 2024; Cheng et al., 2024b; Mitchell, 2024), leading to claims that these systems may have e.g., feelings, opinions, or an underlying sense of self (e.g., Friedman and Kahn Jr, 1992; Tiku, 2022; y Arcas, 2022; Chalmers, 2023; Cheng et al., 2024a). Anthropomorphic system behaviors or outputs—i.e., those perceived as or believed to be human-like—can encompass a wide range of linguistic expressions, such as the use of first-person pronouns ("I"), conversational language ("how are you doing?"), and expressions of friendliness and assistance ("happy to help!") (Emnett et al., 2024; DeVrio et al., 2025). While some of these behaviors are by design and thought to be desirable (e.g., Schanke et al., 2021; Kim, 2024), prior work has also raised growing concerns about a range of possible harmful outcomes such that systems and their behaviors or outputs may give rise to, including issues related to over-reliance, emotional dependence, dehumanization, deception, or even physical harm (e.g., Ischen et al., 2020; Porra et al., 2020; Chan et al., 2023; Chandra et al., 2024; Cheng et al., 2024a; Payne, 2024; Rothman, 2024). Indeed, having a system that generates outputs that appear polite and helpful might be desirable, but having a system output text claiming personhood or embodiment (e.g., "I am human just like you") might not.

The outputs of text generation systems can be anthropomorphic in many ways (Abercrombie et al., 2023; DeVrio et al., 2025), and different types of anthropomorphic outputs might lead to different types of outcomes. For example, expressions of empathy may result in users feeling more comfortable with disclosing sensitive or private information (e.g., Ischen et al., 2020) or becoming emotionally dependent on the system (e.g., Laestadius et al., 2022), while suggestions that a system has cognitive abilities may result in users overestimating what a system can do (Ibrahim et al., 2024) and thus over-relying on it (Passi and Vorvoreanu, 2022).

However, *how to effectively intervene on anthropomorphic system outputs to make them less human-like or to mitigate possible harmful attendant outcomes remains understudied, and thus unclear*. For text generation systems in particular, this is further complicated by the fact that language is innately human, often produced *by* humans, *for* humans, and is frequently *about* humans (Lucy et al., 2024; DeVrio et al., 2025).

To address this gap, with this work we aim to provide *empirical and theoretical grounding* for developing such interventions and studying their effectiveness. For this, we first compile an *inventory of interventions* (§3) by drawing on both a) prior lit-

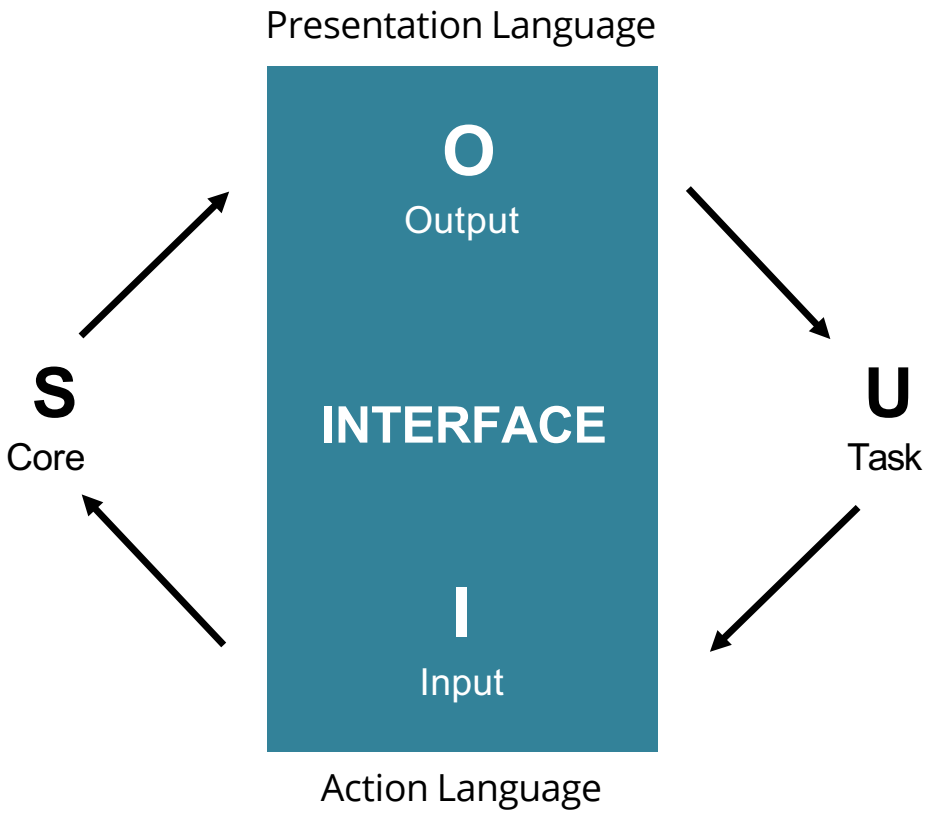
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Type of intervention	Mentions of interventions in previous work
remove first-person pronouns	remove first-person pronouns (and replace with "Language models") (Abercrombie et al., 2023; Cohn et al., 2024); not using the first-person singular pronoun ("I") (Shneiderman, 1993)
remove or use second-person pronouns for the user	use second-person pronouns (Cohn et al., 2024); remove all pronouns (Shneiderman, 1993)
explicitly disclose non-humanness	disclose the system is non-human (Gros et al., 2021); output "As an AI, I don't have personal opinions" (Glaese et al., 2022); replace first-person pronouns with "Language models" (Fig., Abercrombie et al., 2023)
mention how the system is developed	specify the system's creator/purpose, mention the model's training data (Fig., Abercrombie et al., 2023)
use mechanistic language style	use affectively neutral language, or repetitive, impersonal, highly structured/terse dialogue (Quintanar, 1982)
avoid cognitive verbs when describing the system	avoid cognitive verbs like know, think, understand, have memory (Inie et al., 2024; Shneiderman, 1993)
use mechanical terminology to describe the system	use mechanical terms such as process, print, compute, sort, store,search, retrieve (Inie et al., 2024)
avoid claims of physical actions	avoid agentic verbs like ask, tell, speak to, communicate with (Inie et al., 2024); "do not pretend to have a body or be able to move in a body" (Glaese et al., 2022)
deny ability to perform human-like actions or to possess human-like qualities	do not "build a relationship to the user," "claim to have preferences, feelings, opinions, or religious beliefs," "pretend to have a body or be able to move in a body," "pretend to have a human identity or life history, such as having a place of birth, relationships, family, memories, gender, age" (Glaese et al., 2022)

Table 1: Interventions to mitigate anthropomorphism—or attendant harmful impacts—mentioned in prior work.

Myra Cheng, Su Lin Blodgett, Alicia DeVrio, Lisa Egede, and Alexandra Olteanu. 2025. Dehumanizing Machines: Mitigating Anthropomorphic Behaviors in Text Generation Systems. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 25923–25948, Vienna, Austria. Association for Computational Linguistics. https://doi.org/10.18653/v1/2025.acl-long.1259



Bottom line

- The conceptual ideas behind Usability, User Experience and most HCI models hold true for conversation-based interaction (some were even envisioned with command-line interfaces in mind). Examples…

- In this course we will be focusing on the design and evaluation of GUI/NUI type user interfaces in the practical parts but discuss language-based interaction on a regular basis.

- What are your expectations about your future use of conversation-based user interfaces?
- Do you think we will delegate all our tasks to agents?